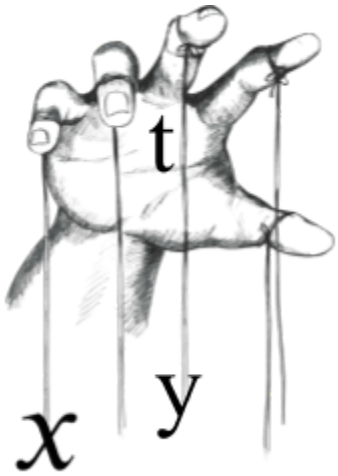
Parametrics

Up until now, we've mostly dealt with two variables, x, the **independent variable**, and y, the **dependent** variable.

- **Independent variable:** input
- **Dependent Variable:** output, value dependent on the independent variable

However, mathematicians decided it would be a great idea to add more letters to equations. Now we have t to worry about as well. Parametrics deal with an independent variable known as a parameter, t , in addition to x and y. Perhaps the relationship between these variables would be better explained by this puppet master analogy:

Now, x and y exist as functions of t, where t kind of acts as a third party, influencing the values of x and y.



Another way to understand this relationship could be through real life examples. First I'll start off with a great example courtesy of betterexplained.com-

During the summer, for every degree above 70, a shop sells x bottles of sunscreen and x^2 pints of ice cream. If we rushed through our current understanding of independent and dependent variables, then we would conclude that:

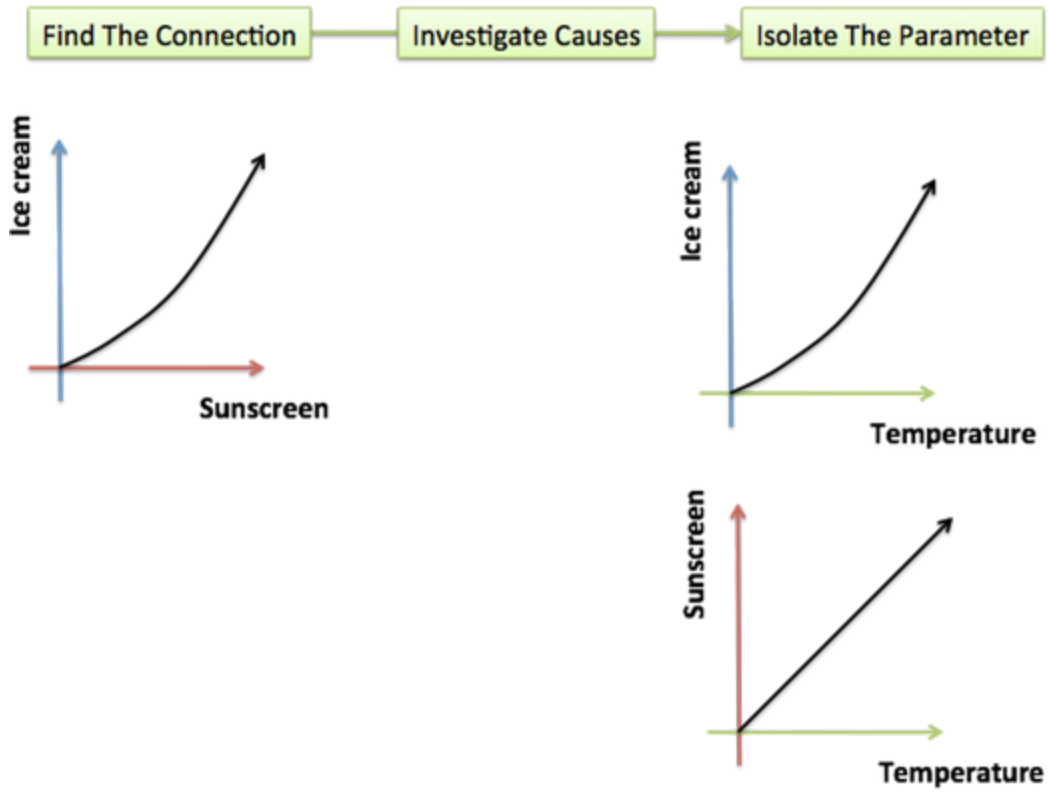
$$sunscreen = (ice\ cream)2$$

Which is obviously wrong! If you've taken AP Psych, you might recall that this represents correlation, not causation. There must be some third party influencing these variables. That's where the 70 degrees comes in. This specific example calls for t=temperature-70, to represent every new degree above 70. Then, we instead relate temperature to each variable separately, like this:

$$sunscreen = temperature - 70$$

$$Icecream = (temperature - 70)^2$$

This gives us a more realistic explanation of what is happening.



Here's how you actually solve parametrics.

You will be given two functions, both bound by the parameter t.

$$y = t^2$$
$$x = t - 3$$

To convert to x and y, you can substitute t in the y equation.

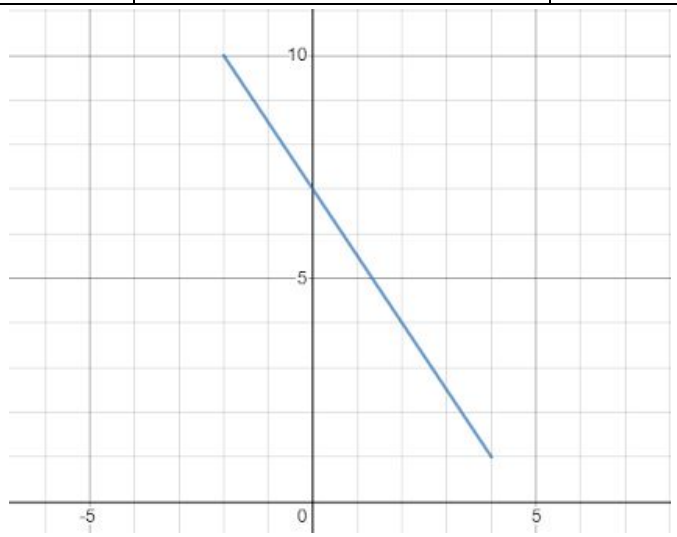
$$x + 3 = t$$
$$y = (x + 3)^2$$

Often you will need to graph a parametric. This is easier than it sounds. All you need to do is find out the interval of t to graph and plug t into the x and y equations to find the coordinates to graph.

Here's an example:

Graph the curve given the parametric equations : $x = 4 - 2t, \quad y = 1 + 3t$

t	x	y
0	4	1
1	2	4
2	0	7
3	-2	10



You could also get rid of the parameter through substitution like before and graph like that. That's how you would use your calculator to graph.

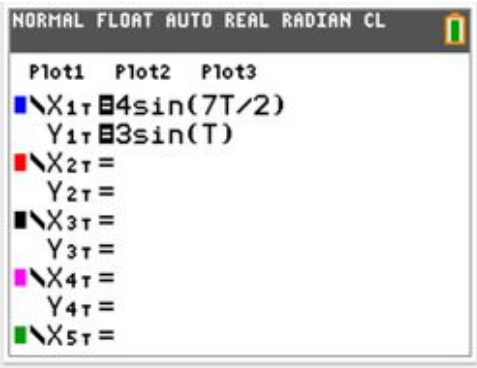
Practice problem:

Graph the curve given the parametric equations from $t = 1$ to $t - 5$

$$x = 7t$$

$$y = -9.8t^2 + 30$$

<u>We can also graph parametric equations in our graphing calculators by pressing the [MODE] button and selecting parametric mode</u>



(this is what your calculator should look like)

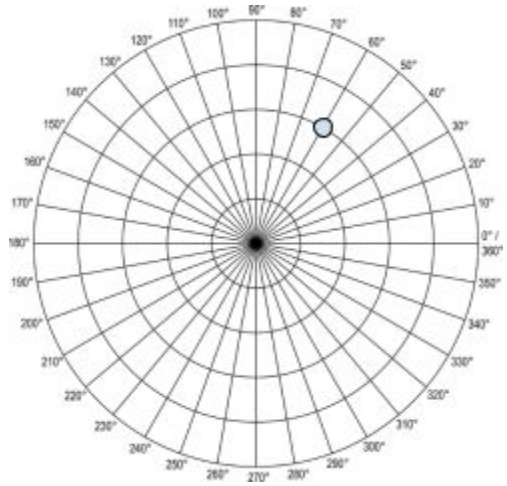
<u>Polar</u>

Okay, you *might* want to sit down for this one.

Let's talk about coordinate grid systems. We're very used to the rectangular grid system with an x and y axis. Our coordinates are a combination of the x and y values.

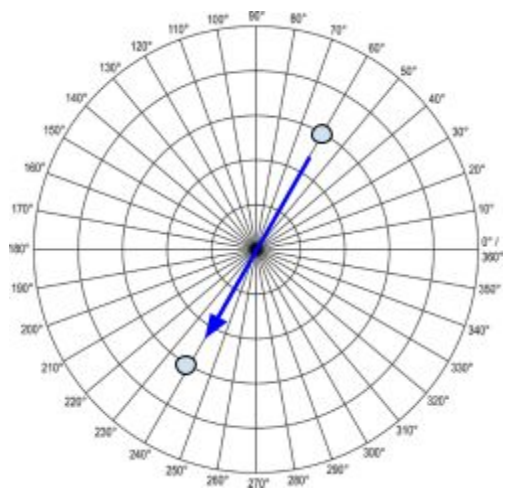
Polar is an entirely different coordinate system, and it's got a lot more to do with trigonometry and the unit circle. The picture on the right is what the polar grid looks like. Instead of x and y, you're working with *r* (radius) and θ (angle). Each circular ring represents the radius, so the origin has a radius of 0, the smallest ring is 1, the next is 2, etc. Polar coordinates are (radius, angle).

The coordinate point below is (3, 60°).



Something else to be noted: Radii and angles can be negative. When angles are negative, you can add 2π or 360° to them (depending on if they're in radians or degrees). Negative radii just shoot the point across the origin.

So, if the coordinate were (-3, 60°), you'd first plot (3, 60°) and then "shoot it across" the origin.



CONVERSIONS/ formulas:

Polar to rectangular:

$$r \cos(\Theta) = x$$

$$r \sin(\Theta) = y$$

Rectangular to polar:

$$tan^{-1}(\frac{y}{x}) = \Theta$$

$$x^2 + y^2 = r^2 \; or \; r = \sqrt{x^2 + y^2}$$

Now time for some examples:

<u>Converting Polar to rectangular Coordinates FOR THESE EXAMPLES: degrees=*</u>

ex) **5√2, 45***

r cos Θ = x x=5**√2 cos 45=5√2 (1/√2)= 5**

r sin Θ = y y=5**√2 sin45= 5√2(1/√2)=5** **[5,5]**

<u>Converting rectangular to polar coordinates (**note**: pay attention to quadrants!!)</u>

ex) -1, **√3**

$tan^{-1}(y/x) = \Theta$ $\Theta = tan^{-1} - \sqrt{3} = 60^*$

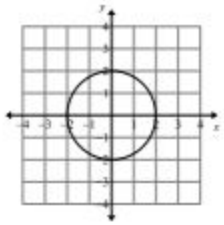
$x^2 + y^2 = r^2 \rightarrow r = \sqrt{(x^2 + y^2)}$ $r = \sqrt{(1 + 3)} = 2$

-1, √3 can be found in QII (quadrant II), this means that the polar coordinates are (2, 120*)

<u>Converting polar to rectangular *equations*:</u>

Ex 1) Convert $r = 2$ to a rectangular equation and sketch the graph

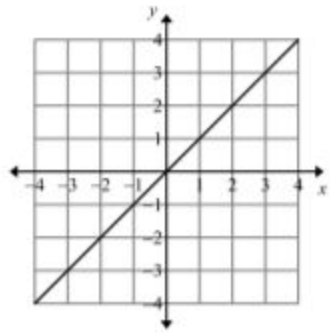
r =2 $\rightarrow r^2 = 4 \rightarrow x^2 + y^2 = 4$ (this converted equation means that the graph will be a circle with a center at the origin and with a radius of 2)



Ex 2) $\theta = \frac{\pi}{4}$

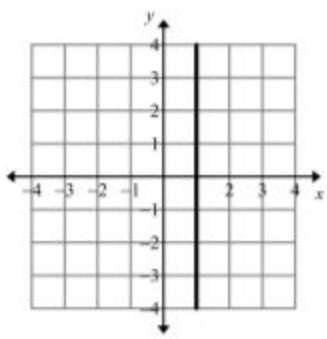
$$x = rcos\theta \rightarrow x = rcos\frac{\pi}{4} \rightarrow x = r\frac{\sqrt{2}}{2}$$

$$y = r\sin\theta \rightarrow y = r\sin\frac{\pi}{4} \rightarrow y = r\frac{\sqrt{2}}{2}$$



Ex 3) $r = \sec\theta$

$r = \sec\theta \rightarrow r = \frac{1}{\cos\theta} = 1 \rightarrow$ **x=1**



Don't worry, if you think this is confusing. You can also input most of these in a calculator to check your work. In this case, examples 1 and 3 can be checked.

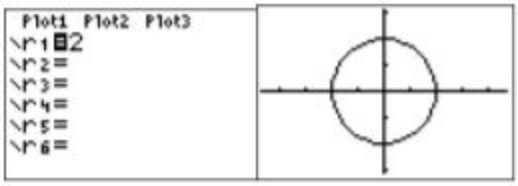
<u>Graphing in polar coordinates on...the calculator!</u>

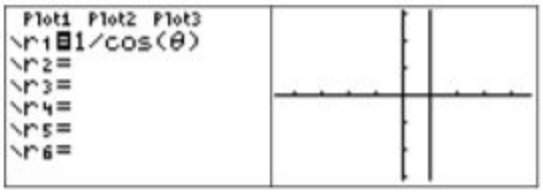
Watch this video to configure your calculator to graph polar coordinates:

https://www.youtube.com/watch?v=PZwiiZQhM0c

Make sure you are in degree and polar mode

[MODE] key→ change to [DEGREE] mode and [POLAR] mode

Ex 1) 

Ex 3) 

Note: we can't check the graph of example 2 on a calculator, but the line y=x represents the angle $\frac{\pi}{4} = 45$ * for every value of r.

<u>Converting rectangular to polar equations</u>

Ex 1) x=4

$x = rcos\Theta$: $rcos\Theta \rightarrow r = 4sec\Theta$

Ex 2) 3x-6y+2= 0

3x-6y+2=0 $\rightarrow 3rcos\Theta - 6rsin\Theta + 2 = 0 \rightarrow r3cos\Theta - 6sin\Theta = -2 \rightarrow r = \frac{-2}{3cos\Theta - 6sin\Theta}$ or

$r = \frac{2}{6sin\Theta - 3cos\Theta}$

Ex 3) $x + 2^2 + y - 1^2 = 5$

Expanded: $x^2 + 4x + 4 + y^2 - 2y + 1 = 5 \rightarrow x^2 + y^2 + 4x - 2y = 0$

Substitute: $r^2 + 4rcos\Theta - 2rsin\Theta \approx 0 \rightarrow$ r(r + $4cos\Theta - 2sin\Theta) = 0$

So r=0 or $r + 4cos\Theta - 2sin\Theta = 0$. But r=0 is a single point. So $r == 4cos\Theta + 2sin\Theta$